

Student Performance Prediction

Introduction

This project focuses on analyzing student performance data and building machine learning models to predict math scores based on academic and demographic features.

The workflow includes data preprocessing, exploratory data analysis (EDA), feature engineering, model training, and performance evaluation using multiple regression models.

The objective of the project is to understand the factors influencing student performance and compare the effectiveness of different machine learning algorithms.

Data Preprocessing

The dataset was initially examined to understand its structure, feature types, and overall data quality before applying machine learning models.

A missing value check was performed to verify whether the dataset contained incomplete or null entries. The analysis showed that the dataset did not contain missing values, allowing the preprocessing workflow to proceed without additional imputation or data cleaning steps.

Categorical features such as gender, race/ethnicity, parental level of education, lunch type, and test preparation course were transformed into numerical format using one-hot encoding. This conversion enabled the machine learning models to process categorical information effectively.

In addition, a new feature named Average Score was created using the mean of math, reading, and writing scores. This feature was used during exploratory analysis to better understand overall student performance trends.

The preprocessing stage ensured that the dataset was structured appropriately for model training and evaluation while maintaining data consistency and interpretability.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to identify meaningful patterns, trends, and relationships within the dataset before model training.

The score distribution analysis showed that most students achieved scores within the mid-to-high range across mathematics, reading, and writing subjects. Reading and writing score distributions appeared relatively similar, while mathematics scores displayed slightly wider variation among students.

Gender-based analysis revealed noticeable performance differences across subjects. Female students generally performed better in reading and writing scores, whereas male students showed comparatively stronger performance in mathematics scores.

The analysis of lunch type indicated that students with standard lunch consistently achieved higher academic scores compared to students receiving free or reduced lunch. This suggests that socioeconomic factors may have an influence on student academic performance.

Parental level of education also showed a visible relationship with student scores. Students whose parents had higher education levels generally achieved better academic performance across subjects.

Correlation analysis demonstrated strong positive relationships between academic subjects, especially between reading and writing scores. This indicates that students performing well in one language-related subject were also likely to perform well in the other.

Model Building

After completing data preprocessing and exploratory analysis, the dataset was divided into training and testing sets to evaluate model performance effectively. An 80:20 train-test split was used for model training and validation.

Three regression models were implemented for predicting student math scores:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Linear Regression was selected as the baseline model because of its simplicity and ability to capture linear relationships between academic scores.

Decision Tree Regressor was implemented to analyze non-linear decision patterns within the dataset. However, single decision trees may sometimes overfit smaller datasets and show reduced generalization performance.

Random Forest Regressor was used as an ensemble learning approach that combines multiple decision trees to improve prediction stability and reduce overfitting.

Results Table

The performance of all models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-Squared Score (R^2).

Among the evaluated models, Linear Regression achieved the best overall performance with the highest R-Squared Score and lowest prediction errors.

Random Forest Regressor also produced strong results and significantly outperformed the Decision Tree Regressor. The Decision Tree model showed comparatively lower performance, which may be due to overfitting and reduced generalization capability.

The results indicate that the dataset contains strong linear relationships between academic features, making Linear Regression highly effective for this prediction task.

Conclusion

This project successfully analyzed student performance data and applied multiple machine learning models to predict math scores using academic and demographic features.

The workflow included data preprocessing, exploratory data analysis, feature engineering, model training, and model evaluation.

The analysis identified strong relationships between reading, writing, and mathematics scores, while factors such as lunch type and parental education also showed visible influence on student performance.

Among the implemented models, Linear Regression achieved the best overall results with the highest R-Squared Score and lowest prediction errors.

Overall, the project demonstrated how machine learning techniques can be effectively applied to educational datasets for performance analysis and prediction.